

COLUMNS

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Introduction

This manual introduces *column sets*, one of the output routines of ConT_EXt. Although column sets are mainly meant for typesetting journals in a semi-automated way, you can also use them for books. We assume that the user is familiar with ConT_EXt and only discuss the commands that are related to column sets. This manual is dedicated to those on the ConT_EXt mailing list who urged me to explain how column sets work.

This mechanism is still under construction which means that functionality may be added and improved. Given the complex nature of this kind of typesetting, you may occasionally run into strange situations.

Column sets are not the most efficient mechanism in ConT_EXt, partly because we're actually dealing with multiple virtual pages (columns) per page, but mostly because of the features that we need to support.

Hans Hagen
PRAGMA ADE
Hasselt, April 2003

Basics

As soon as enough content is collected to build a page, T_EX will invoke the output routine. This is not a fixed piece of code, but a collection of macro calls. The default output routine is a meant for typesetting a single column, as in this document. A multi-column output routine is available as well. This routine mixes well with the single column one, and is activated by:

```
\startcolumns  
some text ...  
\stopcolumns
```

There are some limitations to what you can do with this kind of multi columns, which is why we have a third routine at our disposal: *column sets*. This routine can be used for rather complex layouts with graphics all over the place, and text spanning columns or even spreads. There are of course some shortcomings, which we will discuss later.

In the examples that follow we use the following style to set up the layout:

```
\startenvironment cols-000  
  
\dontcomplain  
  
\useMPlibrary[dum]  
  
\usetypescript  
  [palatino]  
  [\defaultencoding]  
  
\setupbodyfont  
  [palatino]  
  
\setupcolors  
  [state=start]  
  
\setuplayout  
  [grid=yes]  
  
\setuplayout  
  [middle]  
  
\setupsystem  
  [random=1234]  
  
\setuppagenumbering  
  [alternative=doublesided,  
  location=]
```

```
\setupheadertexts
  [pagenumber][right]
  [left][pagenumber]

\setupfootertexts
  [\tttf\inputfilename]

\setuptolerance
  [verytolerant,stretch]

% todo: flatten

\startuniqueMPgraphic{frame}
  fill OverlayBox withcolor transparent(1,.25,red) ;
  fill OverlayBox withcolor transparent(1,.25,blue);
\stopuniqueMPgraphic

\startuniqueMPgraphic{contrast}
  fill OverlayBox withcolor white ;
  fill OverlayBox withcolor transparent(1,.25,red) ;
  fill OverlayBox withcolor transparent(1,.25,yellow) ;
\stopuniqueMPgraphic

\defineoverlay[frame] [\useMPgraphic{frame}]
\defineoverlay[contrast] [\useMPgraphic{contrast}]

\definecolor[red] [r=.75,g=.5,b=.5]
\definecolor[green] [r=.5,g=.75,b=.5]
\definecolor[blue] [r=.5,g=.5,b=.75]

\stopenvironment
```

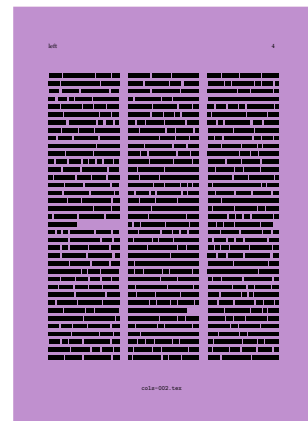
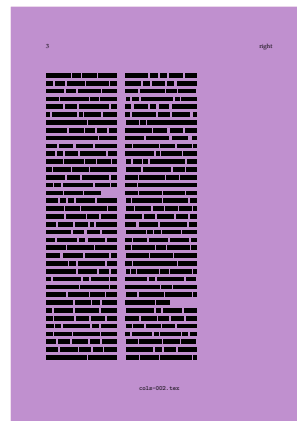
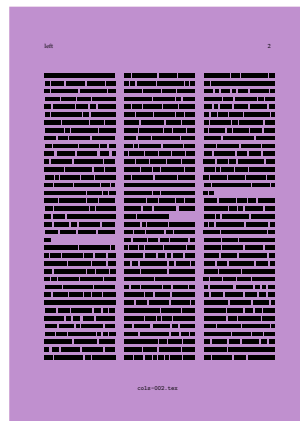
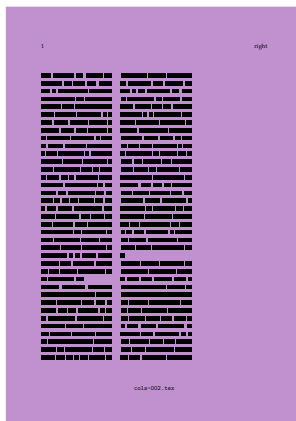
Before we will demonstrate more complex layouts, we will introduce a few features. In the next series of examples we use fake text. You can enlarge the pages in your viewer if you want to see more detail.



These pages were typeset with the following code:

```
\definecolumnset[example][n=2]
\starttext
\startcolumnset[example]
  \dorecurse{15}{\fakewords{100}{200}\par}
\stopcolumnset
\stoptext
```

The number of columns is not fixed to two. You can even have a different number of columns on left and right pages.



This time the input is:

```
\definecolumnset[example][nleft=3,nright=2,width=5cm]
\starttext
\startcolumnset[example]
  \dorecurse{15}{\fakewords{100}{200}\par}
\stopcolumnset
\stoptext
```

In order to get the balancing you want, you can manually influence the height of a column.



When you set the number of column lines to a positive value, that will be the number of lines. A negative value will be subtracted from the maximum number of lines.

```
\definecolumnset [example] [n=2]

\setupcolumnsetlines [example] [1] [1] [-2]
\setupcolumnsetlines [example] [1] [2] [-4]
\setupcolumnsetlines [example] [2] [1] [-6]
\setupcolumnsetlines [example] [2] [2] [-6]
\setupcolumnsetlines [example] [3] [1] [-4]
\setupcolumnsetlines [example] [3] [2] [-2]

\starttext

\startcolumnset [example]

  \dorecurse{15}{\fakewords{100}{200}\par}

\stopcolumnset

\stoptext
```

In articles you may want to let the introduction span multiple columns. A column span is defined independent of a column set and shows up as follows:



Here we've given the span a background so that it stands out.

```
\definecolumnset[example][n=3]
\definecolumnsetspan[wide][n=2,background=contrast]
\starttext
\startcolumnset[example]
  \startcolumnsetspan[wide]
    \dorecurse{2}{\fakewords{25}{50}\par}
  \stopcolumnsetspan
  \dorecurse{15}{\fakewords{100}{200}\par}
\stopcolumnset
\stoptext
```

A column span can be positioned like any graphic. Later we will discuss the placement options in more detail, for the moment all you need to know is that `bt1r` tells ConT_EXt to place the graphic in the left bottom of the the text area.



Here we pass the default placement as parameter to the span, but you can also set it at definition time. We use a brute force simple column splitter to fake columns inside the span.

```
\definecolumnset[example][n=3]
\definecolumnsetspan[wide][n=2,background=contrast]
\starttext
\startcolumnset[example]
  \startcolumnsetspan[wide][default=bt1r]
    \startsimplecolumns
      \dorecurse{2}{\fakewords{25}{40}\par}
    \stopsimplecolumns
  \stopcolumnsetspan
  \dorecurse{15}{\fakewords{100}{200}\par}
\stopcolumnset
\stoptext
```

You are not limited to one column span. In this sense a span is like a graphic: as long as there is room, it will be placed where you want it to be placed. The main difference between a span and a graphic is that a span expects text and tries to align the baselines with the rest of the text. In many ways a column span behaves like a framed text.



This time we flushed one of the spans from bottom to top, starting at the right edge: `btr1`.

```
\definecolumnset[example][n=3]
\definecolumnsetspan[wide][n=2,background=contrast]
\starttext
\startcolumnset[example]
  \startcolumnsetspan[wide]
    \fakewords{25}{50}
  \stopcolumnsetspan
  \fakewords{25}{50}
  \startcolumnsetspan[wide]
    \fakewords{25}{50}
  \stopcolumnsetspan
  \startcolumnsetspan[wide][default=btr1]
    \fakewords{25}{50}
  \stopcolumnsetspan
  \dorecurse{15}{\fakewords{100}{200}\par}
\stopcolumnset
\stoptext
```

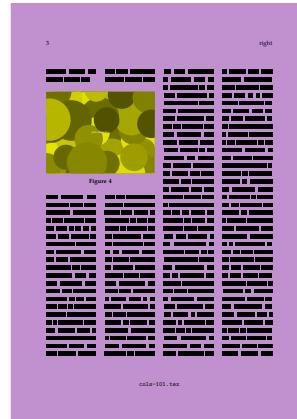
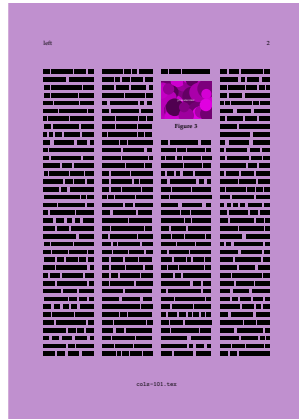
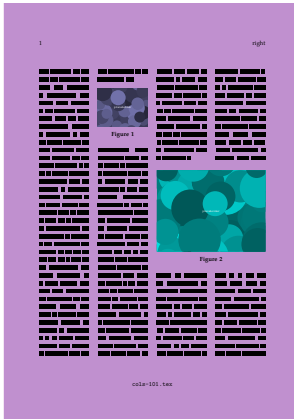
Column spans are treated like graphics, which means that they will float if needed. In the process, the width is limited to the available space, which in some cases may lead to a smaller span than you may expect. Think of a column span, calculated (and prepared) in the last column and ending up on the next page in the first column, where a broader span would have been possible.



```
\definecolumnset[example][n=3]
\definecolumnsetspan[wide][n=2,background=contrast]
\starttext
\startbuffer
  \startcolumnsetspan[wide]
    \fakewords{25}{50}
  \stopcolumnsetspan
  \fakewords{25}{50}
  \startcolumnsetspan[wide]
    \fakewords{25}{50}
  \stopcolumnsetspan
  \startcolumnsetspan[wide][default=btr1]
    \fakewords{25}{50}
  \stopcolumnsetspan
  \dorecurse{3}{\fakewords{100}{200}\par}
\stopbuffer
\startcolumnset[example]
  \dorecurse{4}{\getbuffer}
  \dorecurse{5}{\fakewords{100}{200}\par}
\stopcolumnset
\stoptext
```


Graphics

Most documents have graphics, and therefore column sets can contain them. The main thing that you have to keep in mind when placing graphics, is that column sets are based on grids. Therefore spacing around graphics is also grid based.



```
\definecolumnset[example][n=4]

\starttext

\startbuffer
\fakewords{100}{200}
\placefigure{}{\externalfigure[placeholder][width=\One]}
\fakewords{100}{200}
\placefigure{}{\externalfigure[placeholder][width=\Two]}
\fakewords{100}{200}
\stopbuffer

\startcolumnset[example]

  \definecolumnsethsize{example}{1}{1}\One
  \definecolumnsethsize{example}{1}{2}\Two

  \dorecurse{5}{\getbuffer}
  \dorecurse{10}{\fakewords{100}{200}\par}

\stopcolumnset

\stoptext
```

You can tell ConT_EXt where it should place the graphic, but this will only be honored when there is still place.



The preferred location is passed as a four character directive:

```
\definecolumnset [example] [n=4]

\starttext

\startbuffer
\placefigure[tbrl]      {tbrl}{\externalfigure[fake] [width=\One,  lines=5]}
\placefigure[fxtb:2*4]{fxtb}{\externalfigure[fake] [width=\Two,  lines=5]}
\placefigure[btlr]      {btlr}{\externalfigure[fake] [width=\Two,  lines=5]}
\placefigure[btlr]      {btlr}{\externalfigure[fake] [width=\Three,lines=5]}
\dorecurse{5}{\fakewords{100}{200}\par}
\stopbuffer

\startcolumnset [example]

  \definecolumnsethsize{example}{1}{1}\One
  \definecolumnsethsize{example}{1}{2}\Two
  \definecolumnsethsize{example}{1}{3}\Three

  \dorecurse{4}{\getbuffer}

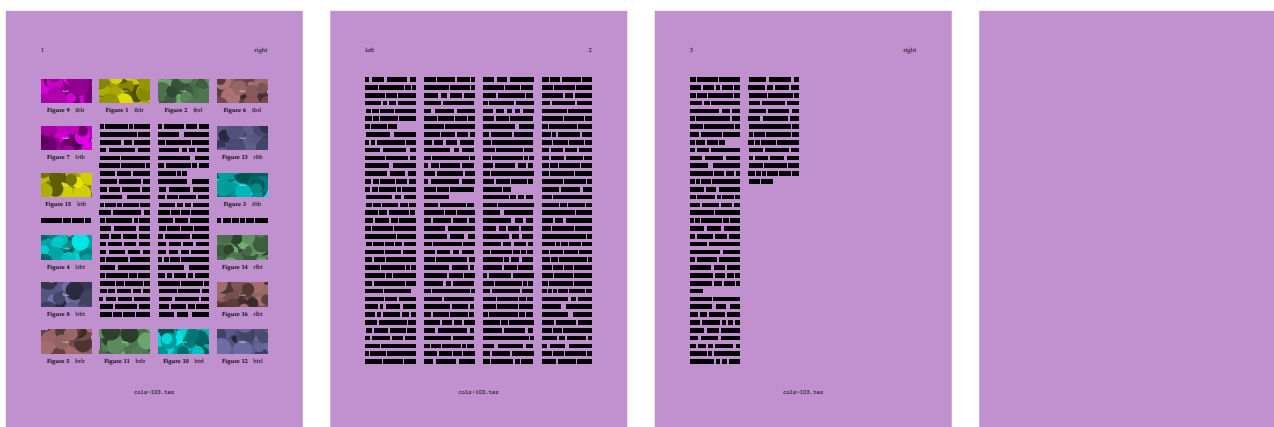
\stopcolumnset
```

`\stoptext`

The following directives are available:

<code>btlr</code>	flush from bottom to top and left to right
<code>btrl</code>	flush from bottom to top and right to left
<code>tblr</code>	flush from top to bottom and left to right
<code>tbrl</code>	flush from top to bottom and right to left
<code>lrbt</code>	flush from left to right and bottom to top
<code>lrtb</code>	flush from left to right and top to bottom
<code>rlbt</code>	flush from right to left and bottom to top
<code>rltb</code>	flush from right to left and top to bottom
<code>here</code>	try to place the graphic here
<code>fixd</code>	force the graphic here and don't float
<code>fxtb:c*r</code>	place the graphic at (c,r) or lower if needed
<code>fxbt:c*r</code>	place the graphic at (c,r) or higher if needed
<code>tops</code>	place the graphic at the top of the page
<code>bots</code>	place the graphic at the bottom of the page
<code>page</code>	place the graphic at a separate page

In the next example we show the 16 directional locations:



`\definecolumnset[example][n=4]`

`\starttext`

`\startcolumnset[example]`

```
\placefigure[btlr]{brrl}{\externalfigure[fake][width=\textwidth,lines=3]}
\placefigure[btrl]{brrl}{\externalfigure[fake][width=\textwidth,lines=3]}
\placefigure[btrl]{btrl}{\externalfigure[fake][width=\textwidth,lines=3]}
\placefigure[tblr]{tblr}{\externalfigure[fake][width=\textwidth,lines=3]}
\placefigure[tblr]{tblr}{\externalfigure[fake][width=\textwidth,lines=3]}
\placefigure[tblr]{tblr}{\externalfigure[fake][width=\textwidth,lines=3]}
```

```
\placefigure[tbrl]{tbrl}{\externalfigure[fake] [width=\textwidth,lines=3]}
\placefigure[tbrl]{tbrl}{\externalfigure[fake] [width=\textwidth,lines=3]}

\placefigure[lrbl]{lrbl}{\externalfigure[fake] [width=\textwidth,lines=3]}
\placefigure[lrbl]{lrbl}{\externalfigure[fake] [width=\textwidth,lines=3]}
\placefigure[lrtb]{lrtb}{\externalfigure[fake] [width=\textwidth,lines=3]}
\placefigure[lrtb]{lrtb}{\externalfigure[fake] [width=\textwidth,lines=3]}
\placefigure[rlbt]{rlbt}{\externalfigure[fake] [width=\textwidth,lines=3]}
\placefigure[rlbt]{rlbt}{\externalfigure[fake] [width=\textwidth,lines=3]}
\placefigure[rltb]{rltb}{\externalfigure[fake] [width=\textwidth,lines=3]}
\placefigure[rltb]{rltb}{\externalfigure[fake] [width=\textwidth,lines=3]}

\dorecurse{10}{\fakewords{50}{100}\par}

\stopcolumnset

\stoptext
```

Areas

So far we have seen texts and graphics that span multiple columns using span commands and floats placement commands. We have also seen that you can define a different number of columns for left and right pages. Now that we have arrived at column areas you will see how we can span information over not only a page but also over pages in a spread.



Being a framed text, by default an area is aligned at the baseline. You can lower an area by setting the `location` to `depth`.

```
\definecolumnset [example] [n=4]

\definecolumnsetarea
  [one] [both]
  [x=1,y=1,nx=2,ny=6,
   background=contrast,state=repeat]

\definecolumnsetarea
  [two] [both]
  [x=4,y=10,nx=1,ny=6,location=depth,
   background=contrast,state=repeat]

\starttext

\startcolumnset [example]

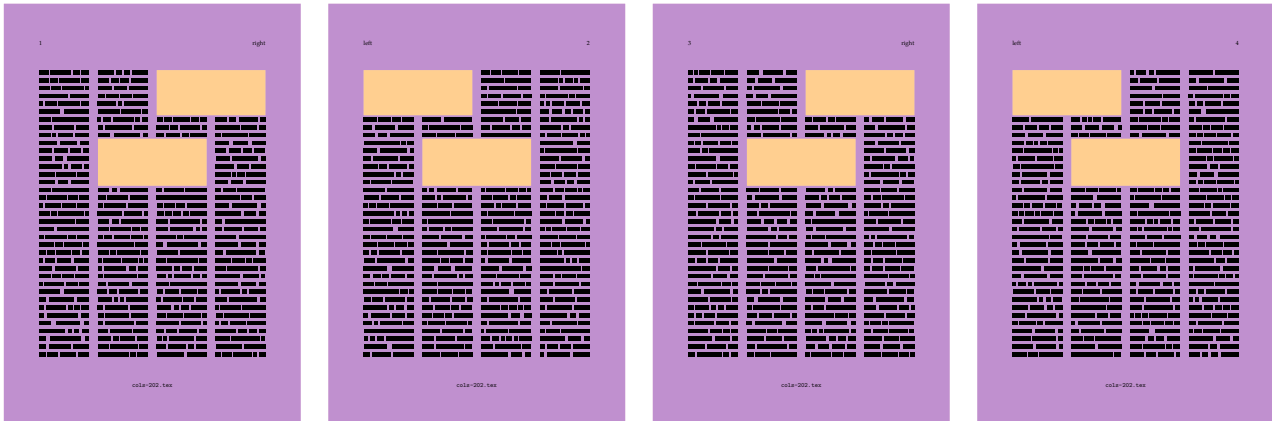
  \dorecurse{30}{\fakewords{50}{100}}

\stopcolumnset

\stoptext
```

Areas

You can position areas on the left, right or next page or on both pages. When you set state to repeat, an area is repeated, otherwise it is only placed once.



Here we just repeat the areas. Normally this only make sense when the content is worth repeating.

```
\definecolumnset[example][n=4]

\definecolumnsetarea
[one][left]
[x=1,y=1,nx=2,ny=6,location=depth,
background=contrast,state=repeat]

\definecolumnsetarea
[two][right]
[x=3,y=1,nx=2,ny=6,location=depth,
background=contrast,state=repeat]

\definecolumnsetarea
[three][both]
[x=2,y=10,nx=2,ny=6,location=depth,
background=contrast,state=repeat]

\starttext

\startcolumnset[example]

\dorecurse{30}{\fakewords{50}{100}}

\stopcolumnset

\stoptext
```

Areas can span a spread, as is demonstrated in the next example.



```
\definecolumnset[example][n=4]

\definecolumnsetarea
  [one][left]
  [x=1,y=1,nx=6,ny=6,
   background=contrast,state=start]

\definecolumnsetarea
  [two][left]
  [x=3,y=10,nx=4,ny=6,location=depth,
   background=contrast,state=start]

\stopsetups

\starttext

\startcolumnset[example]

  \dorecurse{25}{\fakewords{50}{100}}

\stopcolumnset

\stoptext
```

An application of a spread area is a title. In the next example we show two spread pages.



Watch how we explicitly go to a left page.

```
\definecolumnset[example][n=4]

\definecolumnsetarea
  [title][left]
  [x=1,y=1,nx=8,ny=8,
   background=contrast,align={lohi,right},state=start]

\setupcolumnsetareatext
  [title][left]
  [\setups{the title}]

\startsetups[the title]

  \GapText
    {\dimexpr(\makeupwidth+\backspace)} {6pt}
    {RegularBold} {sa 4}
    {A Title Spanning 2 Pages}

\stopsetups

\starttext

\page[left]

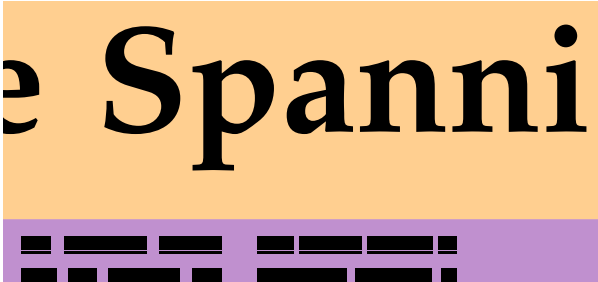
\startcolumnset[example]

  \dorecurse{15}{\fakewords{50}{100}}

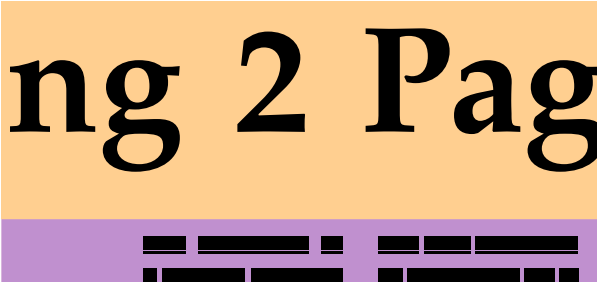
\stopcolumnset

\stoptext
```

The `\GapText` macro is an experimental fun macro and is used to make sure that we don't end up with a clipped character.

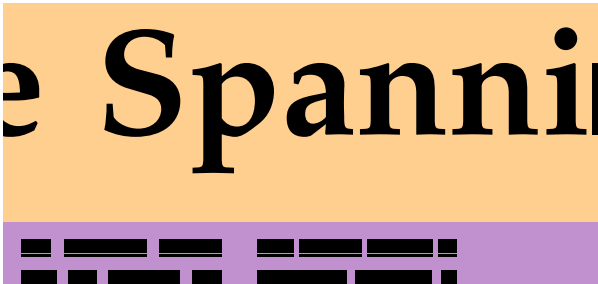


left page

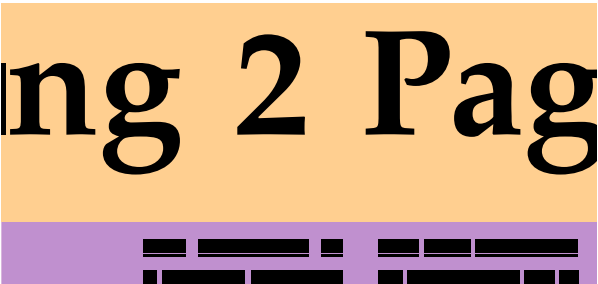


right page

This is a typical example of the kind of optimizations that are needed when you make documents of styles with text that spans a spread. In the next clip we visualize the gap.

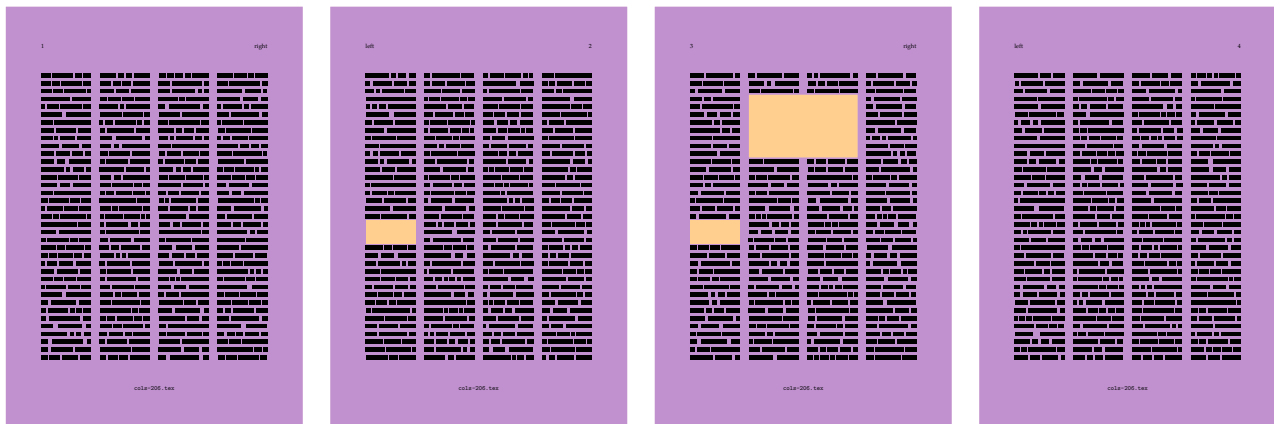


left page



right page

At some moment you may want to set up an area in advance as we have done in the following example.



The page key is used to specify the page in the column set that the area should go into. Column set pages start numbering at 1. The fixed stands for left or right, whatever comes first.

```
\definecolumnset[example][n=4]

\definecolumnsetarea
  [first] [fixed]
  [x=2,y=4,nx=2,ny=8,page=3,location=depth,
   background=contrast,state=start]

\definecolumnsetarea
  [second] [both]
  [x=1,y=20,nx=1,ny=3,page=2,location=depth,
   background=contrast,state=start]

\starttext

\startcolumnset[example]

  \dorecurse{25}{\fakewords{50}{100}}

\stopcolumnset

\stoptext
```

Columns

You can use `\page` to go to a new page in a column set. Likewise you can use `\column` to force a column break.



This example demonstrates that you can supply `\column` with explicit directives.

```
\definecolumnset [example] [n=4]
\starttext
\startcolumnset [example]
  \fakewords{50}{75} \column
  \fakewords{50}{75} \column[yes]   yes:
  \fakewords{50}{75} \column[page] page:
  \fakewords{50}{75} \column[2]     column 2:
  \fakewords{50}{75} \column[3]     column 3:
  \fakewords{50}{75} \column[first] first:
  \fakewords{50}{75} \column[last]  last:
  \fakewords{50}{75} \page          page:
  \fakewords{50}{75} \column[3*5]   3*5:
  \fakewords{50}{75}
\stopcolumnset
\stoptext
```

You can use `\column[page]` as an alternative for `\page`.

Details

This chapter will cover a couple of cosmetic details of column sets. *Some features need to be improved, especially in combination with areas, graphics and alike. We will also provide side floats etc.*



You can set the distance between columns for each pair of columns. *Todo: left and right page distances and margins.*

```
\definecolumnset[example][n=4]
\setupbackgrounds[text][text][background=contrast]
\setupcolumnset[example][2][distance=36pt]
\setupcolumnset[example][3][distance=72pt]
\setupcolumnset[example][4][distance=24pt]
\starttext
\startcolumnset[example]
  \dorecurse{25}{\fakewords{100}{150}\par}
\stopcolumnset
\stoptext
```

Details

When you can the distance of the first column as well. This creates a margin.



This time we used equal distances:

```
\definecolumnset [example] [n=4]
\setupbackgrounds [text] [text] [background=contrast]
\setupcolumnset [example] [distance=24pt]
\setupcolumnset [example] [1] [distance=96pt]
\starttext
\startcolumnset [example]
  \dorecurse {25} {\fakewords {100} {150} \par}
\stopcolumnset
\stoptext
```

The width of columns is normally calculated automatically, but you can also set the width explicitly:



In code:

```
\definecolumnset[example][n=4]
\setupbackgrounds[text][text][background=contrast]
\setupcolumnset[example][width=.15\makeupwidth]
\starttext
\startcolumnset[example]
  \dorecurse{25}{\fakewords{100}{150}\par}
\stopcolumnset
\stoptext
```

For special effects, you can set the width per column. In that case you need to be aware of the fact that T_EX works its way through the document per paragraph. Changing the width halfway a paragraph is possible but will affect the whole paragraph. Therefore, this feature works best when you also goto the next column explicitly.



In code:

```
\definecolumnset [example] [n=4]
\setupbackgrounds [text] [text] [background=contrast]
\setupcolumnset [example] [1] [width=.3\makeupwidth]
\starttext
\startcolumnset [example]
  \dorecurse{25}{\fakewords{50}{75}\column [local]}
\stopcolumnset
\stoptext
```

If you really want to change the width in the middle of a paragraph, you can do a trial run and include a breakpoint at the place where you want it to occur:



Again in code:

```
\definecolumnset [example] [n=4]
\setupbackgrounds [text] [text] [background=contrast]
\setupcolumnset [example] [1] [width=.3\makeupwidth]
\starttext
\startcolumnset [example]
  \fakewords{150}{150}\column [local] \fakewords{25}{50}
  \dorecurse{25}{\fakewords{50}{75}\par}
\stopcolumnset
\stoptext
```

Now, in situations like this, you definitely don't want the second page to have a wide first column as well. You'll want something:



We achieved this by chaining two column sets:

```
\definecolumnset[example][n=4]
\definecolumnset[therest][n=4]

\setupbackgrounds[text][text][background=contrast]

\setupcolumnset[example][1][width=.3\makeupwidth]

\starttext

\startcolumnset[example,therest]
  \fakewords{150}{150}\column[local]\fakewords{25}{50}
  \dorecurse{25}{\fakewords{50}{75}\par}
\stopcolumnset

\stoptext
```

This is still not perfect. Adapting the width to a change in column width is not trivial to implement in T_EX, especially because changes in the output routine do not naturally migrate to the outside world: output routines act in their own grouped environment. However, in most cases such a change is not desirable at all, since the second and following columns need to have equal widths. The following solution is better.



Watch how we explicitly set the widths of the columns before entering the column set.

```
\definecolumnset[example][n=3]
\definecolumnset[therest][n=4]

\setupbackgrounds[text][text][background=contrast]

\setupcolumnset[example][distance=.5cm]
\setupcolumnset[therest][distance=.5cm]

\setupcolumnset[example][width=\dimexpr((\makeupwidth-1.5cm)/4)]
\setupcolumnset[therest][width=\dimexpr((\makeupwidth-1.5cm)/4)]

\setupcolumnset[example][1][width=\dimexpr(\makeupwidth-1.0cm-2
\dimexpr((\makeupwidth-1.5cm)/4))]

\starttext
\startcolumnset[example,therest]
  \startcolumnmakeup
    \vfill \fakewords{50}{100}
  \stopcolumnmakeup
  \column[local]
  \dorecurse{25}{\fakewords{100}{150}\par}
\stopcolumnset
\stoptext
```


Flows

We will not introduce an old feature called text flow, but a bit tricky in combination with column sets:



Beware, this is some experimental code:

```
\definecolumnset [example] [n=4]

\definecolumnsetarea
  [one] [both]
  [x=1,y=1,nx=2,ny=6,
   align=high,offset=1ex,
   background=contrast,state=testflow]

\setupcolumnsetareatext
  [one] [both] [\flushtextflow{demoflow}]

\definetextflow
  [demoflow]
  [style=smallbodyfont,width=\dimexpr(\Two-2ex)]

\definecolumnsethsize{example}{1}{2}\Two

% \definetest[testflow] [next] {\doiftextflowelse{demoflow}}

\definetest[testflow] {\doiftextflowelse{demoflow}}
\starttextflow[demoflow] \stoptextflow % force content

\starttext

\startcolumnset [example]

  \starttextflow[demoflow] \fakewords{100}{125} \stoptextflow

  \dorecurse{30}{\fakewords{50}{100}}
```

\stopcolumnset

\stoptext



Directions

In this chapter we will cover typesetting from right to left as we as chinese.

Backgrounds

As with many ConT_EXt components, column sets can have backgrounds.

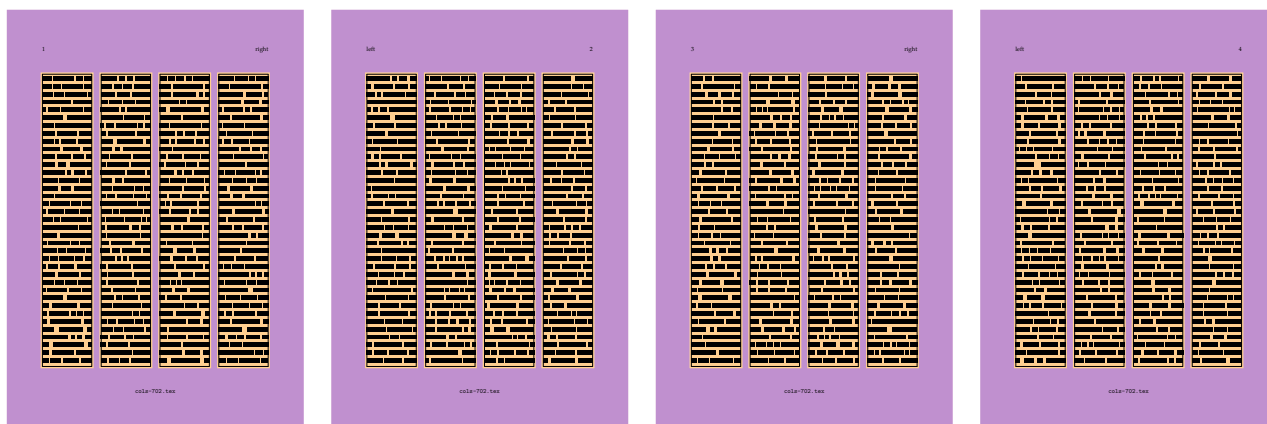


Watch how we use the each keyword to tell ConT_EXt that we want to apply the background to each column of the set.

```
\definecolumnset [example] [n=4]
\setupcolumnset
  [example] [each]
  [background=contrast]
\starttext
\startcolumnset [example]
  \dorecurse{30}{\fakewords{50}{100}}
\stopcolumnset
\stoptext
```

Backgrounds

Normally, if you apply backgrounds this way, you will also set the background offset. In a similar fashion you can apply backgrounds to areas and spans. Such backgrounds can be a color, or any overlay or layer you want.



Here we've set the background offset as well as the frame.

```
\definecolumnset [example] [n=4]

\setupcolumnset
  [example] [each]
  [background=contrast,
   backgroundoffset=3pt,
   frame=on,
   rulethickness=1pt]

\starttext

\startcolumnset [example]

  \dorecurse{30}{\fakewords{50}{100}}

\stopcolumnset

\stoptext
```

When dealing with areas, you need to be aware of the fact that they are clipped, the content as well as the background.



The default clip offset is two times the lineheight, except in the binding, where it is set to zero points. You can set the clip offset with the `clipoffset` parameter.

```
\definecolumnset [example] [n=4]

\definecolumnsetarea
  [title] [left]
  [x=1,y=1,nx=8,ny=8,
   background=contrast,state=repeat]

\setupcolumnsetareatext
  [title] [left] [\setups{cow}]

\startsetups [cow]

  \externalfigure
    [cow.pdf]
    [height=15\lineheight,
     width=2\dimexpr(\makeupwidth+\backspace)]

\stopsetups

\starttext

\startcolumnset [example]

  \dorecurse{30}{\fakewords{50}{100}}

\stopcolumnset

\stoptext
```

The text background mechanism is rather well adapted to column sets. The following example is a variant on an example shown in the manual titled details.



Watch how we adapt the background to the fact and extent that the text spans one or more columns.

```
\definecolor[shadecolor][r=.5,g=.5,b=.25] % yellow

\startuseMPgraphic{mpos:par:columnset:shade}
  path p, q, r ; numeric h ;
  for i=1 upto nofmultipars :
    p := multipars[i] ;
    h := bbheight(p) ;
    q := multipars[i] topenlarged -.5h ;
    r := multipars[i] bottomenlarged -.5h ;
    if one_piece_multi_par :
      linear_shade(q,8,white,boxfillcolor) ;
      linear_shade(r,8,boxfillcolor,white) ;
    elseif multilocs[i] = 1 :
      linear_shade(p,8,boxfillcolor,white) ;
    elseif multilocs[i] = 2 :
      linear_shade(q,8,boxfillcolor,white) ;
      linear_shade(r,8,white,boxfillcolor) ;
```

```

    else :
        linear_shade(p,8,white,boxfillcolor) ;
    fi ;
endfor ;
\stopuseMPgraphic

\definecolumnset
[example]
[n=4]

\definetextbackground
[shade]
[location=paragraph,
backgroundcolor=shadecolor,
mp=mpos:par:columnset:shade,
method=mpos:par:columnset,
leftoffset=\topskipgap,
before=\blank,
after=\blank]

\starttext

\startcolumnset[example]

\dorecurse{40}
{\starttextbackground[shade]
\fakewords{10}{200}
\stoptextbackground}

\stopcolumnset

\stoptext

```

Backgrounds that follow the paragraph shape also work ok in column sets.

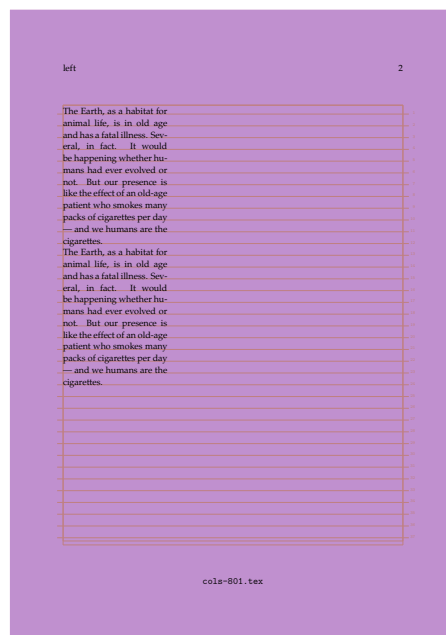
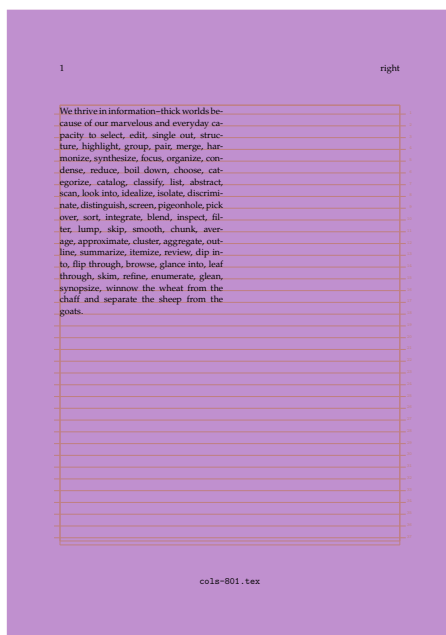
todo: an example of a bleeding graphic with column feed back (from techniek)



Spanning and More

todo: explanation

Spanning and More

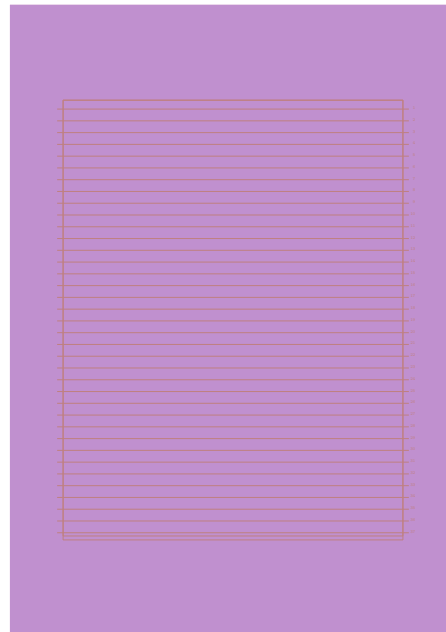
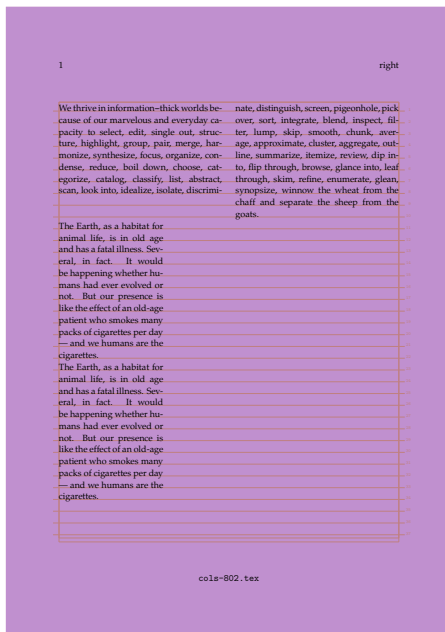


```
\definecolumnset [example-1] [n=2]
\definecolumnset [example-2] [n=3]

\starttext \showgrid

  \startcolumnset [example-1] \dorecurse {1}{\input tufte \par} \stopcolumnset
  \startcolumnset [example-2] \dorecurse {2}{\input ward \par} \stopcolumnset

\stoptext
```



```

\definecolumnset [example-1] [n=2,balance=yes]
\definecolumnset [example-2] [n=3,balance=yes]

\starttext \showgrid

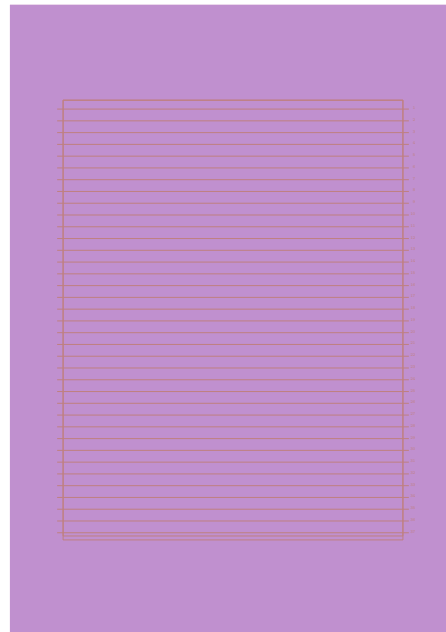
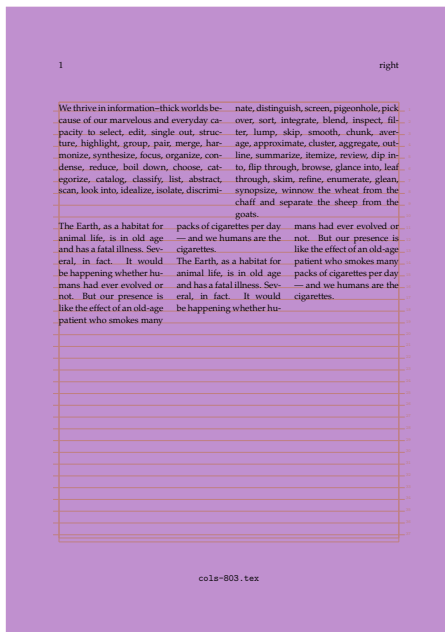
\setupcolumnsetlines[example-1][1][1] [8]
\setupcolumnsetlines[example-1][1][2] [10]

\startcolumnset [example-1] \dorecurse {1}{\input tufte \par} \stopcolumnset
\startcolumnset [example-2] \dorecurse {2}{\input ward \par} \stopcolumnset

\stoptext

```

Spanning and More



```
\definecolumnset [example-1] [n=2,balance=yes]
\definecolumnset [example-2] [n=3,balance=yes]

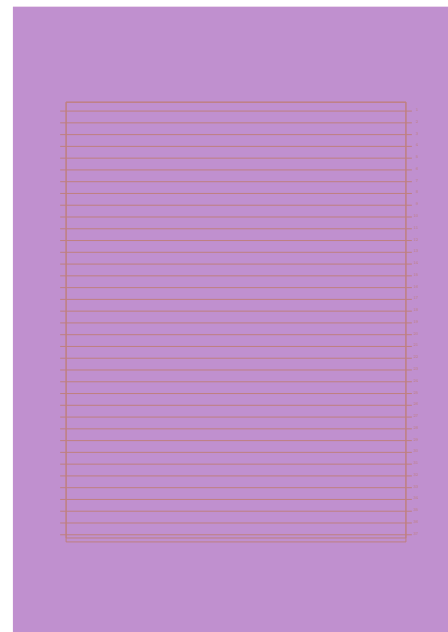
\starttext \showgrid

\setupcolumnsetlines[example-1][1][1][8]
\setupcolumnsetlines[example-1][1][2][10]

\setupcolumnsetlines[example-2][1][1][19]
\setupcolumnsetlines[example-2][1][2][18]
\setupcolumnsetlines[example-2][1][3][19]

\startcolumnset [example-1] \dorecurse {1}{\input tufte \par} \stopcolumnset
\startcolumnset [example-2] \dorecurse {2}{\input ward \par} \stopcolumnset

\stoptext
```



```

\definecolumnset [example-1] [n=2,balance=yes]
\definecolumnset [example-2] [n=3,balance=yes]

\starttext \showgrid

\setupcolumnsetlines[example-1][1][1] [8]
\setupcolumnsetlines[example-1][1][2] [10]

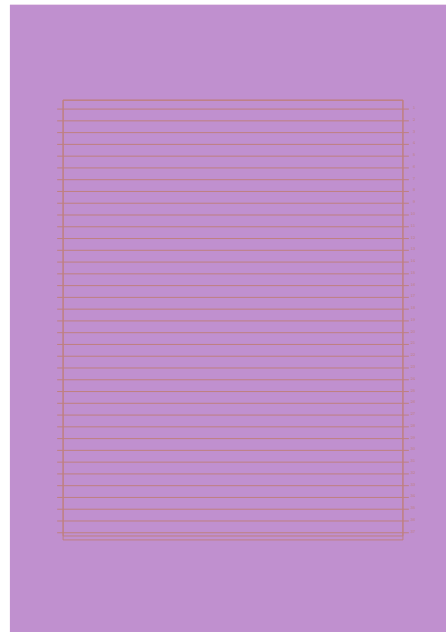
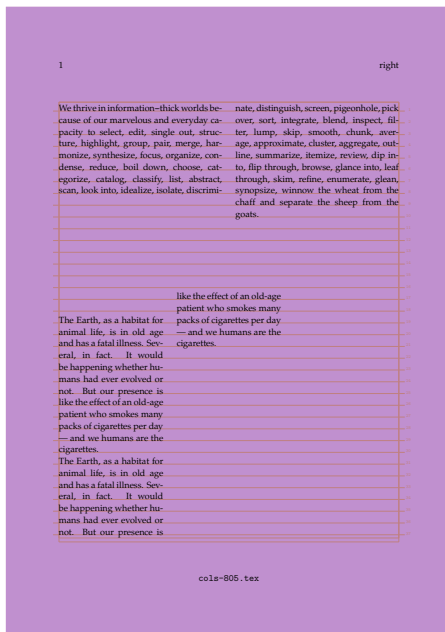
\setupcolumnsetlines[example-2][1][1] [0]
\setupcolumnsetlines[example-2][1][2] [-2]
\setupcolumnsetlines[example-2][1][3] [0]

\startcolumnset [example-1] \dorecurse {1}{\input tufte \par} \stopcolumnset
\startcolumnset [example-2] \dorecurse {6}{\input ward \par} \stopcolumnset

\stoptext

```

Spanning and More



```
\definecolumnset [example-1] [n=2,balance=yes]
\definecolumnset [example-2] [n=3,balance=yes]

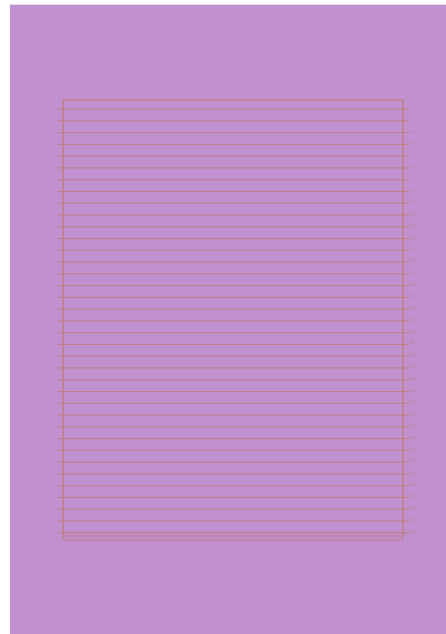
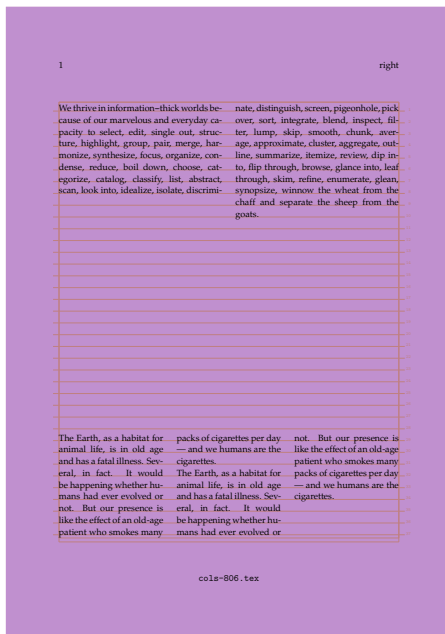
\starttext \showgrid

\setupcolumnsetlines[example-1][1][1] [8]
\setupcolumnsetlines[example-1][1][2] [10]

\setupcolumnsetstart[example-2][1][1] [19]
\setupcolumnsetstart[example-2][1][2] [17]
\setupcolumnsetstart[example-2][1][3] [19]

\startcolumnset [example-1] \dorecurse {1}{\input tufte \par} \stopcolumnset
\startcolumnset [example-2] \dorecurse {2}{\input ward \par} \stopcolumnset

\stoptext
```



```

\definecolumnset [example-1] [n=2,balance=yes]
\definecolumnset [example-2] [n=3,balance=yes]

\starttext \showgrid

\setupcolumnsetlines[example-1][1][1] [8]
\setupcolumnsetlines[example-1][1][2] [10]

\setupcolumnsetstart[example-2][1][1] [29]
\setupcolumnsetstart[example-2][1][2] [29]
\setupcolumnsetstart[example-2][1][3] [29]

\startcolumnset [example-1] \dorecurse {1}{\input tufte \par} \stopcolumnset
\startcolumnset [example-2] \dorecurse {2}{\input ward \par} \stopcolumnset

\stoptext

```

